IOS APPLICATION DEVELOPMENT

COURSEWORK PROPOSAL

PantryPulse

Smart Food Inventory, Recognition and Expiry Management iOS Application

Module	iOS Application Development
Programme	BSc (Hons) Computing
Index Number	COBSCCOMP25.1P - 029
Assessment	Coursework Proposal
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1. App Idea, Target Audience and Unique Selling Points

PantryPulse is a native iOS application designed to help users manage household food inventory, monitor expiry dates and reduce avoidable food waste. Users can add food manually, scan packaged products using barcodes, or photograph common foods for on-device AI recognition. The app stores food information locally, tracks expiry dates, sends reminders and creates a prioritised 'Use First' list so users can quickly see which food needs attention.

The target audience includes household users, students and young adults who want a simple way to keep track of groceries and avoid forgetting food. The main USP is the combination of practical pantry management with two meaningful on-device ML capabilities: Vision + Core ML food recognition and Core ML waste-risk prediction. The AI is integrated into the application's core workflow rather than being a separate chatbot or decorative feature.

2. Problem Statement

Users can lose track of food stored at home, forget expiry dates, purchase duplicate items and fail to consume food before it spoils. PantryPulse addresses these problems through a visual digital pantry, expiry intelligence and actionable prioritisation.

- Forgetting expiry dates.
- Limited visibility of food already stored at home.
- Duplicate grocery purchases.
- Difficulty deciding which food should be consumed first.
- Time-consuming manual entry of packaged products.
- Lack of personalised awareness of foods likely to be wasted.

3. Feature Descriptions

The following nine MVP features form the core application, while the final two are the advanced on-device AI/ML features planned for the advanced iOS SDK component.

1. Smart Dashboard

Shows total pantry items, expiring-soon items, urgent items and recently added food, giving users an immediate overview instead of requiring them to search through the inventory.

2. Pantry Inventory Management

Allows users to add, edit, delete and view food items with name, category, quantity, purchase date, expiry date, storage location, image and notes.

3. Barcode Scanning

Uses the iPhone camera to detect packaged-food barcodes, reducing the amount of information the user has to enter manually.

4. External Product API Integration

Sends a detected barcode to a genuine product-information REST API and displays the returned product information for user confirmation. This provides the required external API/backend integration.

5. Core Data Persistence

Stores pantry items and consumption/waste records locally so data remains available across app launches and can be used by the application's business logic.

6. Expiry Tracking and Alerts

Calculates remaining shelf life, identifies items approaching expiry and schedules local notifications to reduce forgotten food.

7. Use First

Creates a prioritised list of items that should be consumed first, primarily using expiry information and enhanced by waste-risk predictions.

8. Consumption and Waste Tracking

Lets users mark food as consumed or wasted. These events provide useful history for insights and the waste-risk ML feature.

9. Food Waste Insights

Displays statistics such as consumed, wasted and saved items and commonly wasted categories, helping users understand and improve their food-management behaviour.

10. Advanced – AI Food Recognition

The user photographs a food item. Vision processes the image and passes it to an on-device Core ML image-classification model. The result may be shown as 'Banana – 96% confidence' and is confirmed by the user before being added to the pantry.

11. Advanced – ML Waste-Risk Prediction

A second Core ML model predicts Low, Medium or High waste risk using structured inputs such as food category, quantity, days until expiry, storage type and consumption/waste history. The prediction affects the Use First list and recommendations.

4. Advanced AI Workflows

4.1 AI Food Recognition

Camera → Vision Framework → Core ML Image Classifier → Food + Confidence → User Confirmation → Core Data

The recognition model will be scoped to a manageable set of common food categories. If confidence is low, the user can correct the result manually. The model runs on-device and does not depend on an external AI/LLM.

4.2 ML Waste-Risk Prediction

Food data + expiry + quantity + history → Core ML Waste-Risk Model → Low / Medium / High → Use First / Recommendation

The waste-risk model is separate from image recognition. It uses structured food and usage information. For new users, general features such as category, quantity and days until expiry can be used; as the user records consumed and wasted items, their history can provide additional inputs.

5. Tools and Services Justification

Tool / Service	Justification
Swift	Native iOS development language with direct access to Apple SDKs and strong type safety.
SwiftUI	Modern declarative UI framework for building the application's screens, reusable components and navigation.
Xcode	Apple's development environment for building, debugging, testing and integrating iOS frameworks.
Core Data	Provides persistent local storage for pantry, consumption and waste records and supports the mandatory local-storage MVP requirement.
Vision Framework	Provides camera-based image analysis and barcode detection and forms the bridge between camera input and Core ML.
Core ML	Runs the food image-classification model and waste-risk prediction model entirely on-device, making the advanced AI functionality self-contained.
UserNotifications	Provides local expiry reminders so users receive alerts without requiring a remote notification server.
WidgetKit	Provides a Home Screen widget showing urgent/Use First food items at a glance.
App Intents	Can expose selected PantryPulse actions or information to Siri and Shortcuts as an additional system integration if development time permits.
External Product API	A genuine food/product-information REST API such as Open Food Facts, or another approved equivalent, will provide product information from scanned barcodes and satisfy the external API requirement.
Swift Testing	Used to test expiry logic, inventory operations, prioritisation, API handling and ML-related input preparation.

6. Coursework MVP and Advanced SDK Alignment

- Core Data will provide persistent local storage.
- Local notifications will provide expiry reminders.
- A genuine external product-information API will provide barcode-based product lookup.
- Vision + Core ML will provide on-device AI food recognition.
- Core ML will provide the separate waste-risk prediction feature.
- The advanced features are integrated into core application logic rather than being superficial SDK demonstrations.

7. Main User Flows

7.1 AI Food Recognition

1. Open AI Scan.
2. Capture a food image.
3. Vision processes the image.
4. Core ML predicts the food category and confidence.
5. User confirms or corrects the result.
6. User confirms quantity and expiry information.

7. The item is saved to Core Data.

7.2 Barcode Product Flow

8. Open Barcode Scan.
9. Vision detects the barcode.
10. The barcode is sent to the external product API.
11. Product information is returned.
12. User confirms or edits the information.
13. The product is saved to Core Data.

7.3 Waste-Risk Flow

14. Food is added to the pantry.
15. Relevant food and usage features are prepared.
16. The Core ML waste-risk model predicts Low, Medium or High risk.
17. The prediction is displayed in Food Details.
18. The result contributes to Use First prioritisation and recommendations.

8. Application Architecture

The application will use a modular MVVM-oriented structure, separating the SwiftUI presentation layer from application logic, persistence, ML services and the external API service.

SwiftUI → ViewModels → Core Data / ML Services / API Service → Application Logic → Notifications / WidgetKit

9. UI/UX and Screen Plan

- Onboarding / Welcome
- AI Recognition Introduction
- Home Dashboard
- Pantry Inventory
- AI Food Scanner
- AI Recognition Result
- Barcode Scanner
- Product API Result
- Manual Add Food
- Food Details
- Use First
- Waste-Risk Details
- Food Waste Insights
- Notifications
- Profile / Settings
- Home Screen Widget

10. One-Month Development Plan

Period	Planned Work

Week 1	SwiftUI structure, navigation, dashboard, pantry CRUD, Core Data and base UI.
Week 2	Vision barcode scanning, external product API, product confirmation and expiry logic.
Week 3	Vision + Core ML food recognition and Core ML waste-risk prediction.
Week 4	Notifications, Use First refinement, insights, WidgetKit if feasible, testing, accessibility, UI polish and final documentation.

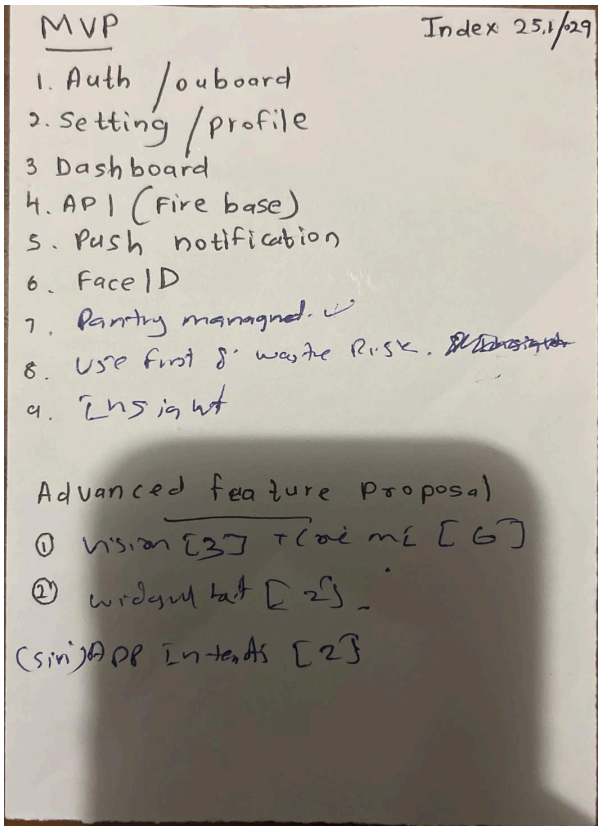
11. Testing and Risk Management

- Test expiry calculations and boundary dates.
- Test Core Data persistence and inventory operations.
- Test API success, missing product and network-failure cases.
- Test low-confidence AI predictions and manual correction.
- Test different waste-risk input combinations.
- Test notification scheduling.
- Test empty pantry, expired items and missing product information.
- Keep manual entry as a fallback for AI recognition and API lookup.

12. Expected Outcome

The final outcome will be a functional native iOS application that combines a practical food-inventory workflow with genuine external integration and meaningful on-device AI. Users will be able to manage food, identify products, monitor expiry, receive reminders, prioritise items and understand their food-waste patterns.

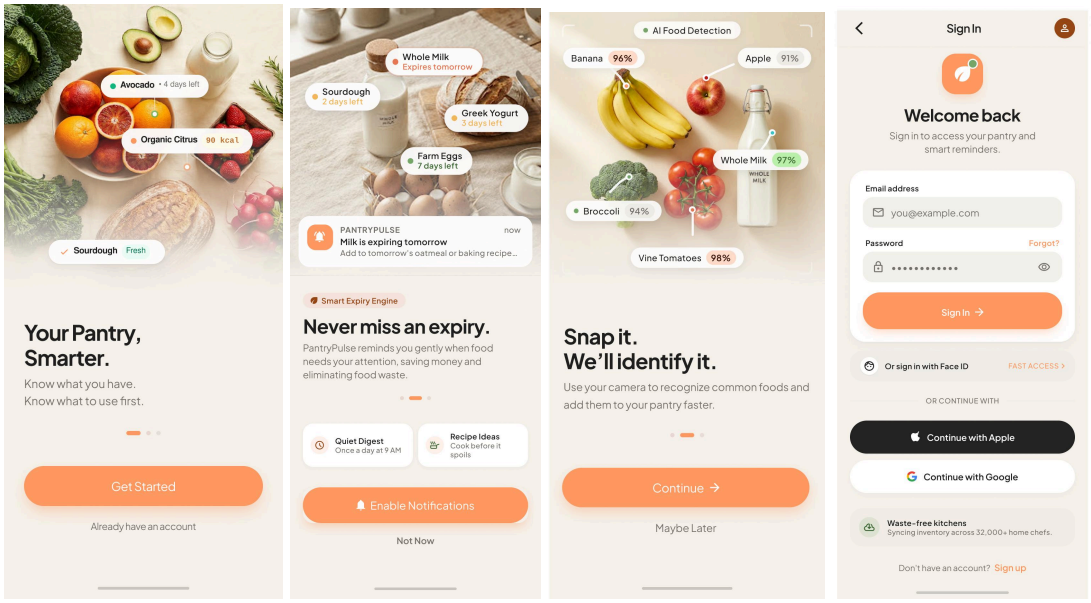
13. MVP form

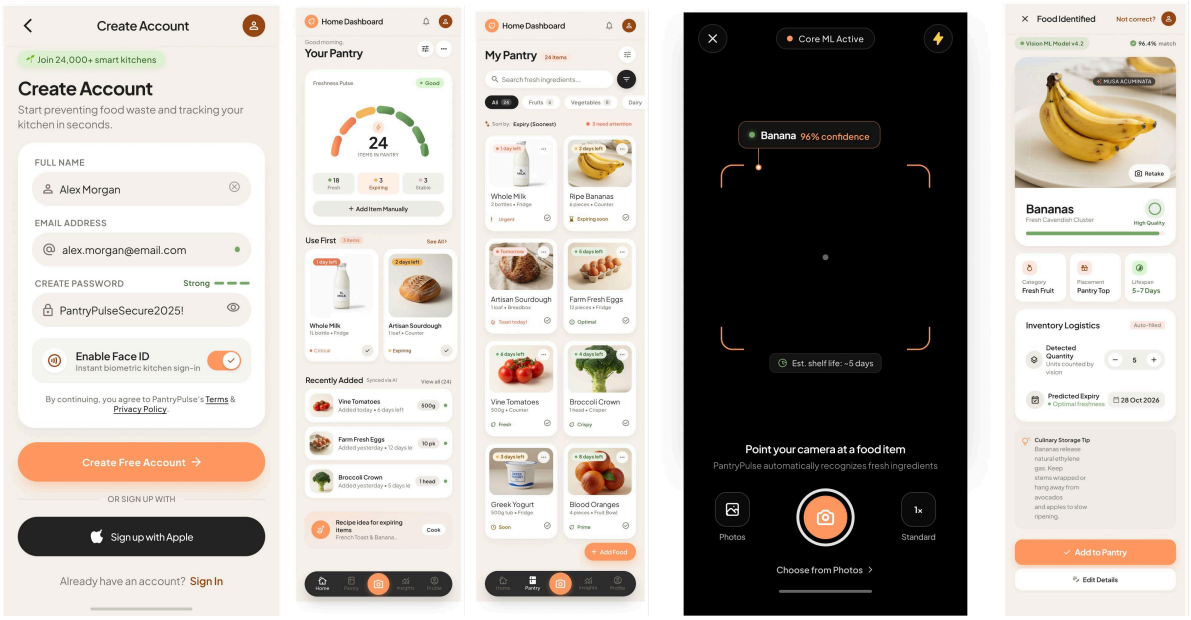


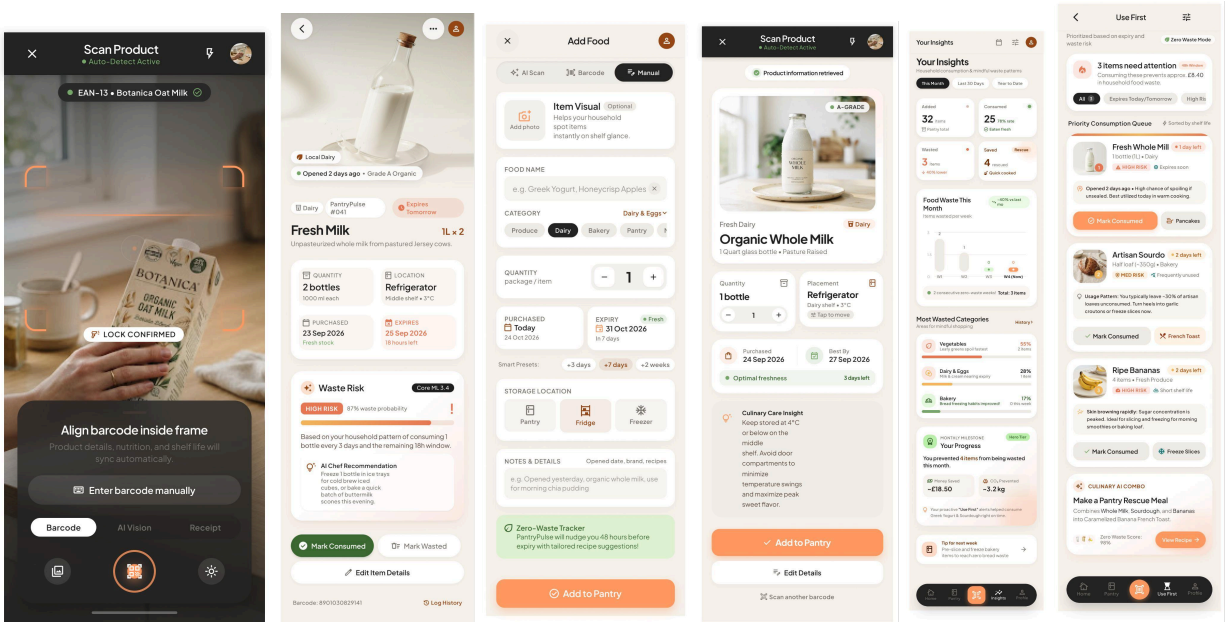
14. UI Designs

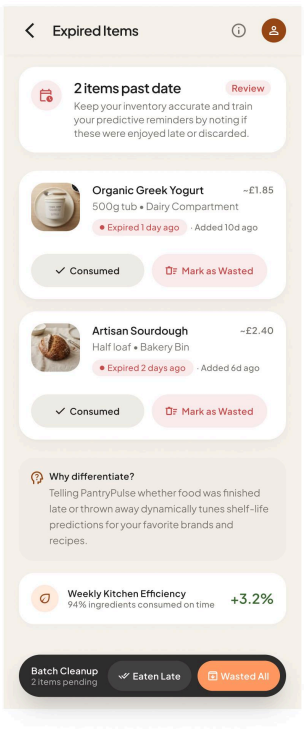
Figma link -

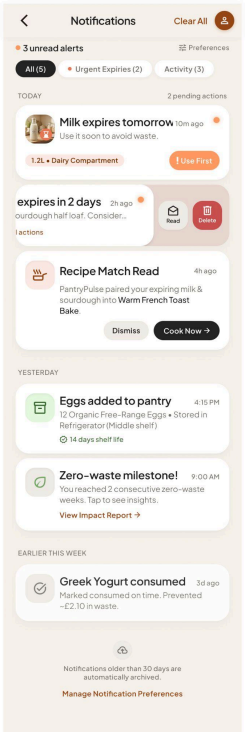
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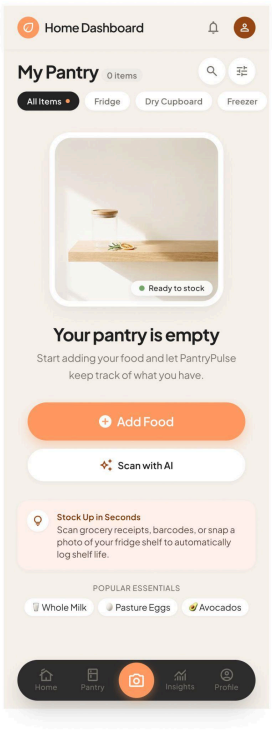

Avocado • 4 days left
Organic Citrus 90 kcal
Sourdough Fresh
Your Pantry, Smarter.
Know what you have.
Know what to use first.
Get Started
Already have an account
Whole Milk Expires tomorrow
Sourdough 2 days left
Greek Yogurt 3 days left
Farm Eggs 7 days left
PANTRYPULSE now
Milk is expiring tomorrow
Add to tomorrow's oatmeal or baking recipe...
Smart Expiry Engine
Never miss an expiry.
PantryPulse reminds you gently when food needs your attention, saving money and eliminating food waste.
Quiet Digest
Once a day at 9 AM
Recipe Ideas
Cook before it spoils
Enable Notifications
Not Now
AI Food Detection
Banana 96%
Apple 91%
Whole Milk 97%
Broccoli 94%
Vine Tomatoes 98%
Snap it.
We'll identify it.
Use your camera to recognize common foods and add them to your pantry faster.
Continue
Maybe Later
Sign In
Welcome back
Sign in to access your pantry and smart reminders.
Email address
you@example.com
Password
Forgot?
Sign In
Or sign in with Face ID
FAST ACCESS
OR CONTINUE WITH
Continue with Apple
Continue with Google
Waste-free kitchens
Syncing inventory across 32,000+ home chefs.
Don't have an account? Sign up

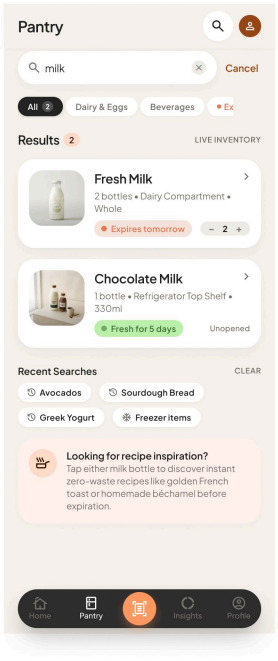

Create Account
Join 24,000+ smart kitchens
Create Account
Start preventing food waste and tracking your kitchen in seconds.
FULL NAME
Alex Morgan
EMAIL ADDRESS
alex.morgan@email.com
CREATE PASSWORD
Strong
PantryPulseSecure2025!
Enable Face ID
Instant biometric kitchen sign-in
By continuing, you agree to PantryPulse's Terms & Privacy Policy.
Create Free Account
OR SIGN UP WITH
Sign up with Apple
Already have an account? Sign In
Home Dashboard
Your Pantry
24
ITEMS IN PANTRY
Add Item Manually
Use First
Recently Added
Vine Tomatoes
Farm Fresh Eggs
Broccoli Crown
Recipe idea for expiring items
My Pantry
Whole Milk
Ripe Bananas
Artisan Sourdough
Farm Fresh Eggs
Vine Tomatoes
Broccoli Crown
Greek Yogurt
Blood Oranges
Core ML Active
Banana 96% confidence
Est. shelf life: ~5 days
Point your camera at a food item
PantryPulse automatically recognizes fresh ingredients
Photos
Standard
Choose from Photos
Food Identified
Not correct?
Bananas
Retake
Inventory Logistics
Detected Quantity
Predicted Expiry
28 Oct 2026
Add to Pantry
Edit Details

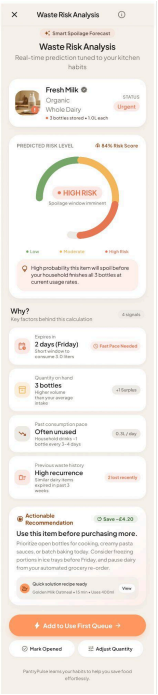

Scan Product
Auto-Detect Active
EAN-13 • Botanica Oat Milk
LOCK CONFIRMED
Align barcode inside frame
Product details, nutrition, and shelf life will sync automatically.
Enter barcode manually
Barcode
AI Vision
Receipt
Local Dairy
Opened 2 days ago • Grade A Organic
Dairy
PantryPulse #041
Expires Tomorrow
Fresh Milk
1L × 2
Unpasteurized whole milk from pastured Jersey cows.
QUANTITY
2 bottles
1000 ml each
LOCATION
Refrigerator
Middle shelf • 3°C
PURCHASED
23 Sep 2026
Fresh stock
EXPIRES
25 Sep 2026
18 hours left
Waste Risk
Core ML 3.4
HIGH RISK
87% waste probability
Based on your household pattern of consuming 1 bottle every 3 days and the remaining 18h window.
AI Chef Recommendation
Freeze 1 bottle in ice trays for cold brew iced cubes, or bake a quick batch of buttermilk scones this evening.
Mark Consumed
Mark Wasted
Edit Item Details
Barcode: 8901030825141
Log History
Add Food
AI Scan
Barcode
Manual
Item Visual
Optional
Add photo
Helps your household spot items instantly on shelf glance.
FOOD NAME
e.g. Greek Yogurt, Honeycrisp Apples
CATEGORY
Dairy & Eggs
Produce
Dairy
Bakery
Pantry
QUANTITY
package / item
1
PURCHASED
Today
24 Oct 2026
EXPIRY
Fresh
31 Oct 2026
In 7 days
Smart Presets:
+3 days
+7 days
+2 weeks
STORAGE LOCATION
Pantry
Fridge
Freezer
NOTES & DETAILS
Opened date, brand, recipes
e.g. Opened yesterday, organic whole milk, use for morning chia pudding.
Zero-Waste Tracker
PantryPulse will nudge you 48 hours before expiry with tailored recipe suggestions!
Add to Pantry
Scan Product
Auto-Detect Active
Product information retrieved
A-GRADE
Fresh Dairy
Dairy
Organic Whole Milk
1 Quart glass bottle • Pasture Raised
Quantity
1 bottle
Placement
Refrigerator
Dairy shelf • 3°C
Tap to move
Purchased
24 Sep 2026
Best By
27 Sep 2026
Optimal freshness
3 days left
Culinary Care Insight
Keep stored at 4°C or below on the middle shelf. Avoid door compartments to minimize temperature swings and maximize peak sweet flavor.
Add to Pantry
Edit Details
Scan another barcode
Your Insights
This Month
Last 30 Days
Year to Date
Added
32
Consumed
25
Wasted
3
Saved
4
Food Waste This Month
Most Wasted Categories
Vegetables
Dairy & Eggs
Bakery
Your Progress
You prevented 4 items from being wasted this month.
Money Saved
~£18.90
CO2 Prevented
~3.2 kg
Use First
3 items need attention
Expires Today/Tomorrow
High Rs
Priority Consumption Queue
Sorted by shelf life
Fresh Whole Milk
1 bottle (1 L) • Dairy
HIGH RISK
Expires soon
Mark Consumed
Pancakes
Artisan Sourdo
2 days left
Half loaf (~ 350g) • Bakery
HIGH RISK
Frequently unused
Mark Consumed
French Toast
Ripe Bananas
2 days left
4 items • Fresh Produce
HIGH RISK
Short shelf life
Mark Consumed
Freeze Slices
CULINARY AI COMBO
Make a Pantry Rescue Meal
Combines Whole Milk, Sourdough, and Bananas into Caramelised Banana French Toast.
View Recipe

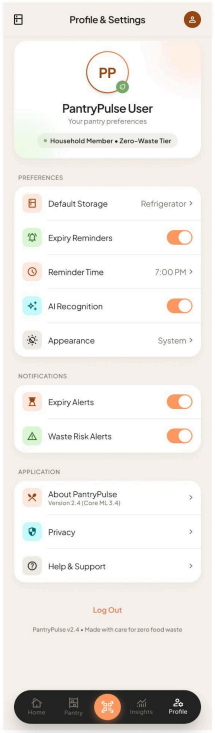


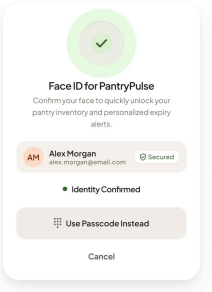












15. Conclusion

PantryPulse provides a clear user problem, a focused MVP and two technically meaningful advanced features. The proposal is intentionally scoped for an individual development period while ensuring that the advanced SDK technologies are deeply connected to the application's core purpose.